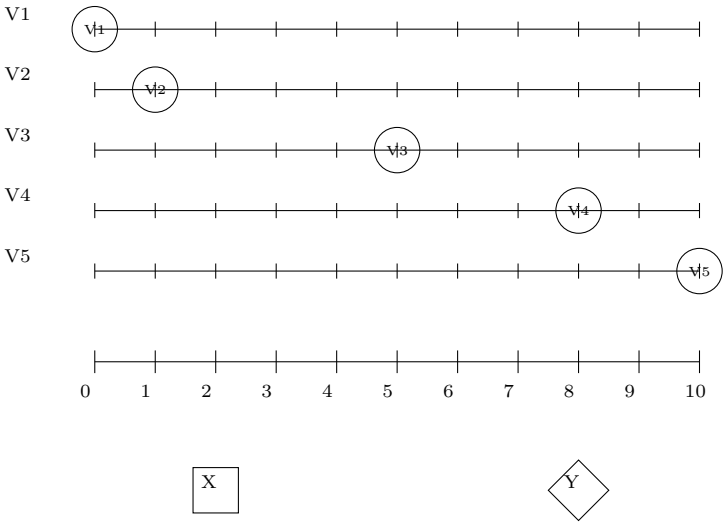
Each favorite marker is one voter. Favorites stay still. Each voter chooses the nearer proposal, X or Y. Equal distances put that vote in Tie. More votes wins; equal vote counts give a tied vote. Use only the numbered ticks.

Fixed favorites; proposals X=2, Y=8



Distances to X / Y; choice

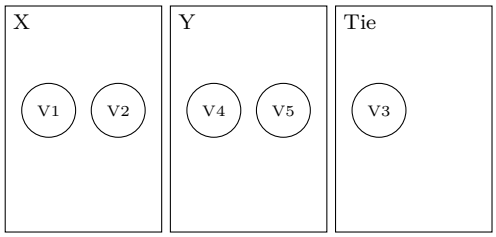
V1: 2 / 8 → X

V2: 1 / 7 → X

V3: 3 / 3 → Tie

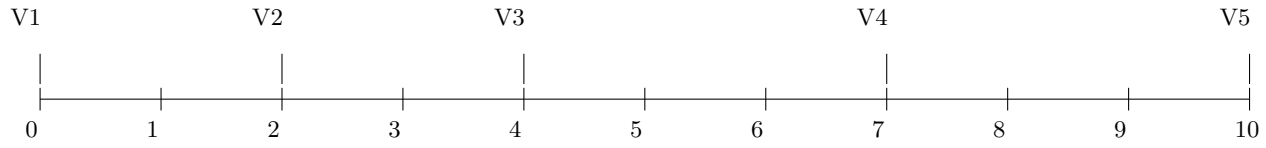
V4: 6 / 0 → Y

V5: 8 / 2 → Y



Tied vote: 2 to 2; V3 in Tie.

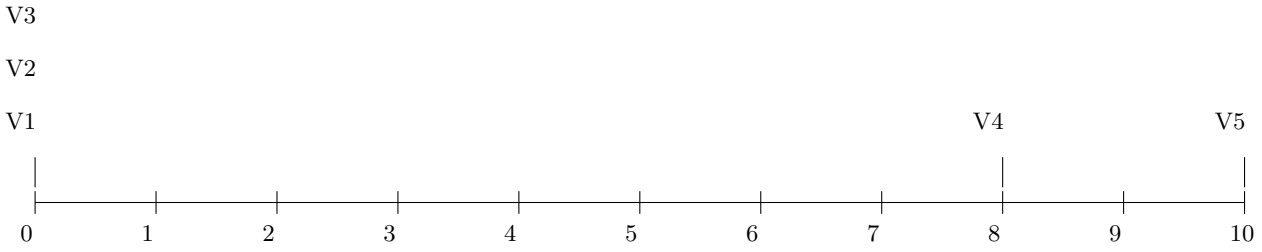
Problem 1: Use these five favorites. Find a location that wins a vote against every different location on the line.



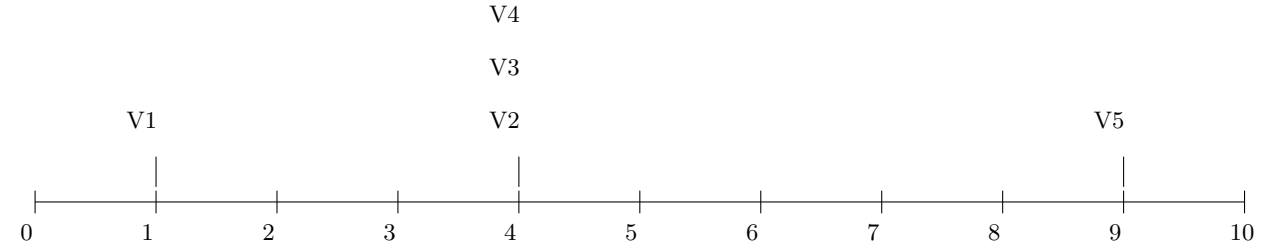
Your location:

Problem 2: Find a location that beats every other location for each set of five favorites. Voters sharing a favorite still get one vote each.

Winning location: _____



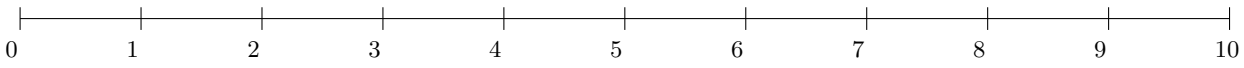
Winning location: _____



Winning location: _____



Problem 3: Put five favorites anywhere on the numbered ticks. Count all five markers, even when some share a place. How can you tell which location beats every other location without trying every vote? Explain why your way always works.



Problem 4: Use just four voters, with these favorites. Find every location that never loses against another location. Is there any location that beats every other one?

